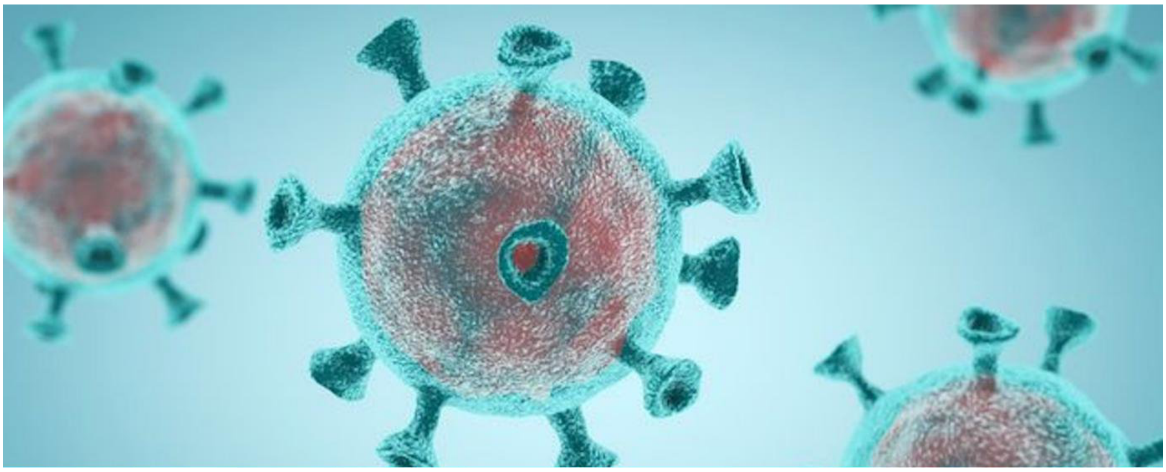
Machine Learning for Design



Assignment 2 Group:
G 4
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Introduction

During the covid pandemic the shift to remote learning impacted the student experience at the TU Delft. In this report we are challenged with improving this covid-era experience, our group used text processing and machine learning techniques to analyse the anonymous student feedback. Our analysis is ranging from VADER sentiment analysis to topic modelling and revealed a returning pattern of feeling of isolation. Students frequently highlighted "online lectures," "working from home," and a significant "lack of social contact" as their primary obstacles during this covid-era, leading to a measurable decline in mental well-being and academic drive.

Solving this problem is critical, since the pandemic did more than just transfer the lectures to online. It took away the social aspect of learning. Our sentiment analysis showed a negative trend, where even seemingly positive comments often asked “needs for support”. The lack of motivation and peer connection doesn’t just hurt grades, it also leads to a disconnected campus culture where the collaborative aspect is lost.

To end this gap we propose the TUD Social Study Hub. This solution consists of a digital matching platform that pairs students based on courses and interests to work together. Also physical study spaces, this are zones on campus specially designed for students to create a collaborative growth. And lastly autonomous networking, this is a feature that allows students to build their own academic social circles.

During the covid pandemic students couldn’t come to the faculty, which meant they had to study from their home. The students at the TU Delft for example were asked to help with a project to improve the bachelors students’ experience during the covid-era. Students had to share their experience anonymously and tell what the main problem was they encountered. With text processing skill we as a group analysed the text to find ways to improve the students’ experience in these difficult times.

Once our solutions are used on campus, we expect the student experience to shift from isolation to collaborative growth. The success of this solution will be seen back in future data, where our models will see clusters like: “community”, “support” and “together”. Ultimately
students won’t be sitting behind their desk all day, but will actually feel part of a connected campus, whether they are studying at home or at TU Delft.

Text analysis

Text preprocessing

To ensure the analysis is reliable, a preprocessing pipeline was used. The pre-processing stage is very important, as machine learning models require structured and numerical input: raw text, however, is unstructured and cannot be processed directly. This is because the text data cannot be processed by ML models in one go, as it consists of unstructured text rather than numerical features. The aim of preprocessing is therefore to reduce noise and convert the text into more comparable and meaningful tokens. The preprocessing pipeline consists of the following steps:

1) **Lowercasing**: all text has been converted to lower case to ensure consistency, so that “Home” and “home” are treated as the same word. This results in less redundancy and reduces the dimensionality of the dataset.
2) **Tokenisation**: the sentences have been split into individual words, or tokens, allowing the text to be analysed at word level, which is very important for topic and frequency analysis.
3) **Removal of punctuation and numbers**: numbers and punctuation have been removed, as they do not contribute to meaningful semantic information and also because they introduce noise into the analysis.
4) **Stopword removal**: standard English stopwords, such as “and”, “the” and “is”, have been removed using a predefined stopword list from the NLP library. These words occur frequently but convey little meaning, which can distort the analysis.
5) **Custom stopwords**: additional domain-specific words have also been removed, such as “TU”, “student”, “study” and “Delft”, as these words occur very frequently in the dataset but do not contribute to distinguishing between different topics, thereby altering the results.
6) **Stemming and lemmatisation**: the words have been reduced to their base form, such as “studying” to “study”, through lemmatisation and stemming. This reduces the variation in word forms and does the same for sparsity in the dataset used. This results in a more robust analysis, whilst it may also reduce nuance.

Implementation

The preprocessing was executed using the following function:

```
tokens = [
preprocess(sentence, lower=True,
word_tokenization=True,
rem_punc=True, rem_numb=True,
rem_stopwords=True,
extra_stopwords=["people",
"go","becaus","n't","realli","student","tu","delft","university","study","studies","day","year","time","week","period", "veri","peopl","studi","lot","littl","feel","problem","difficult","thing","teacher"],
stem=True,
lem=True) for sentence in corpus
]
print(tokens)
```

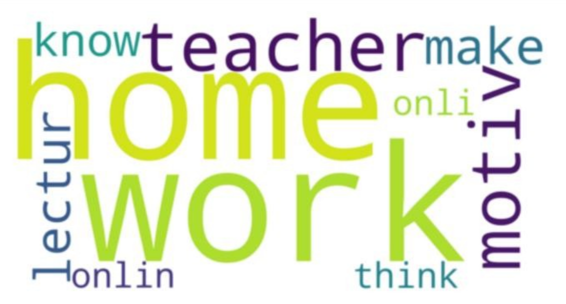
The function described above is based on the implementation from the course tutorial and utilises standard NLP techniques, such as tokenisation, stop-word removal, lemmatisation and stemming.

Reflection on pre-processing

The pre-processing has significantly improved the dataset by standardising words and removing noise. One potential limitation is that the combination of stemming and lemmatisation can result in a reduction of semantic nuances. However, this trade-off is accepted, as the result leads to clearer recognition of patterns in the data.

Word Frequencies Analysis

The first exploratory step involved a word frequency analysis, which was carried out to identify the most frequently occurring words in the dataset.



Following preprocessing, words such as “*work*”, “*home*”, “*teacher*”, “*lecture*” and “*motivation*” are more clearly visible. Initially, the most frequent words consisted mainly of stop words. Using these insights, the pre-processing pipeline was fine-tuned by expanding the list of stop words and applying lemmatisation and stemming. This adjustment improved the quality of the results and led to a more meaningful distribution of the words.

Interpretation: these words show that students mainly talk about the following at home:

- Studying from home
- Online learning
- Problems with motivation

In the context of COVID-19, this indicates that home study and online learning are key topics.

Limitation: the word frequency analysis treats these words as independent of one another and therefore does not take into account the context or relationships between the words. As a result, this method provides only a simplified representation of the actual data.

Sentiment Analysis

After doing the word frequency analysis we started the sentiment analysis, in which we used VADER to analyse the amount of negative, positive and neutral sentiments in the provided dataset.

VADER was chosen because this model has been specifically designed for shorter, more informal texts and also because it produces good results with subjective data, such as student feedback. Furthermore, it generates interpretable scores, such as negative, neutral and positive scores as well as a composite score.

VADER assigns sentiment scores to each text based on a predefined lexicon of words that are associated with positive or negative emotions. Each sentence receives a positive, neutral, and negative score, as well as a compound score that represents the overall sentiment of the statement. The breakdown of the sentiment scores is as follows: negative (280), neutral (108) and positive (233). From this analysis we found that the statements can mainly be divided into two groups. Most of the responses contain a negative sentiment, which indicates that many students describe problems or difficulties. However, there are also a considerable number of positive statements, while neutral statements occur less frequently.

When examining the positive statements more closely, it becomes clear that some of them contain a double meaning. For example, the sentence “would love TU Delft could help find good psychologist” receives a positive sentiment score because of the phrase “would love”. However, the actual meaning of the sentence is more neutral or even slightly negative, because it expresses a need for psychological support. This illustrates the concept of semantic ambiguity, where certain words can carry a positive sentiment in the lexicon but represent a different meaning in the context of the sentence. This shows that VADER provides a useful general overview of sentiment, although it can misinterpret context-dependent meanings. Even with this limitation, the sentiment analysis clearly shows that negative experiences dominate the dataset.

```
sentiment
negative    280
positive    233
neutral     108
Name: count, dtype: int64
```

Frequency of sentiment analysis being positive, negative or neutral.

```
                                          sentences  neg    neu    pos  \
526                  realli like questionair hope help  0.0  0.198  0.802
254                        would like free coffe campus  0.0  0.341  0.659
161                    would interest result well check  0.0  0.370  0.630
584                 better workplac cours home veri nice  0.0  0.412  0.588
580  would love tu delft could help find good psych...  0.0  0.419  0.581

     compound sentiment
526    0.7964  positive
254    0.7003  positive
161    0.6249  positive
584    0.6908  positive
580    0.9100  positive
```

Table with most positive sentiment analysis.

```
                                           sentences    neg    neu  pos  \
598               loneli panic attack hopeless frustrat  0.825  0.175  0.0
534  anxieti stress lack motiv concentr problem fatigu  0.661  0.339  0.0
607  loneli boredom stress fear see famili angri en...  0.655  0.345  0.0
355  concentr problem lack social interact poor qua...  0.618  0.382  0.0
590  veri sick corona veri tire aros psychic proble...  0.605  0.395  0.0

     compound sentiment
598   -0.8555  negative
534   -0.7783  negative
607   -0.8860  negative
355   -0.7964  negative
590   -0.8481  negative
```

Table with most negative sentiment analysis.

```
                                         sentences  neg  neu  pos  \
11             wait time tu delft psychologist decreas  0.0  1.0  0.0
441              daytim activ outsid door bond univers  0.0  1.0  0.0
445  communic especi longer period onli digit reall...  0.0  1.0  0.0
444                        littl motiv studi home lot  0.0  1.0  0.0
436               tuition fee reduc durat corona crisi  0.0  1.0  0.0

     compound sentiment
11        0.0   neutral
441       0.0   neutral
445       0.0   neutral
444       0.0   neutral
436       0.0   neutral
```

Table with most neutral sentiment analysis.

TOPIC MODELING

```
[Topic 0]: 0.011*"also" + 0.011*"work" + 0.009*"would" + 0.008*"onlin" + 0.007*"get" + 0.007*"think" + 0.007*"could"
[Topic 1]: 0.016*"home" + 0.014*"contact" + 0.013*"work" + 0.011*"less" + 0.010*"lectur" + 0.010*"physic" + 0.009*"get"
[Topic 2]: 0.012*"would" + 0.011*"like" + 0.010*"onli" + 0.008*"work" + 0.008*"make" + 0.008*"think" + 0.007*"also"
[Topic 3]: 0.017*"also" + 0.014*"work" + 0.011*"onlin" + 0.010*"contact" + 0.008*"make" + 0.008*"group" + 0.008*"home"
[Topic 4]: 0.010*"also" + 0.010*"onlin" + 0.009*"get" + 0.009*"exam" + 0.009*"much" + 0.008*"place" + 0.008*"home"
[Topic 5]: 0.014*"contact" + 0.012*"fellow" + 0.010*"know" + 0.009*"also" + 0.008*"stress" + 0.007*"ask" + 0.007*"home"
```

Topic modeling can be seen as a more advanced form of word frequency analysis. Instead of only counting how often individual words appear, topic modeling analyzes how often words occur together within the same responses. By analyzing these word co-occurrences, the model groups frequently related words into topics that represent common themes in the dataset. The decision to use topic modelling was made because, unlike word frequency analysis, it can identify relationships between words and makes it possible to identify underlying themes in the data.

After running the model, we identified three topics. Each topic consists of a set of words that frequently appear together in the student responses. The three topics were chosen because this provided a clear and easily explainable distinction between themes, whilst avoiding excessive overlap between different topics.

[Topic 0]: 0.012*"would" + 0.009*"also" + 0.009*"get" + 0.008*"work" + 0.008*"make" + 0.008*"lectur" + 0.007*"contact" + 0.006*"fellow" + 0.006*"think" + 0.006*"onlin"

[Topic 1]: 0.011*"also" + 0.009*"work" + 0.009*"onlin" + 0.009*"home" + 0.007*"make" + 0.007*"much" + 0.007*"get" + 0.007*"think" + 0.007*"like" + 0.006*"exam"

[Topic 2]: 0.014*"contact" + 0.011*"work" + 0.011*"also" + 0.009*"home" + 0.007*"much" + 0.007*"onlin" + 0.007*"fellow" + 0.007*"social" + 0.007*"like" + 0.006*"less"

When interpreting these topics, several recurring themes become visible. Many of the words are related to working online and studying from home, such as online, home, and lecture. In addition, words such as contact, fellow, and social appear frequently, which suggests that students often mention the lack of social interaction with other students.

Overall, these topics indicate that the main themes in the dataset are related to online education from home and the lack of social contact between students.

These findings are consistent with the results from the word frequency analysis (e.g., “home”, “online”, “work”) and the sentiment analysis, where negative emotions dominate. This consistency across multiple methods increases the reliability of the results. By combining word frequency analysis, sentiment analysis and topic modelling, we ensured that both the surface-level patterns and the underlying relationships within the data were captured.

Students’ Input

From the text analysis, we identified three important student needs that frequently appear in the dataset. The first important need is more social contact between students. In the topic modelling results, words such as contact, fellow, and social frequently appeared together. Based on this, we conclude that students often indicate that they experience a lack of social interaction with fellow students and that this is perceived as unpleasant. This is also visible in several responses from the dataset. Some students mention that they struggle with the lack of social contact during their studies. An example of this is: “*The lack of social contacts, I came to live new in Delft myself and I have not been able to meet new people yet.*”

A second important need mentioned by students relates to online education and studying from home. In the word frequency analysis, words such as lecture, home, work, and online appeared frequently. Many students describe that it is difficult to actively participate in online lectures and indicate that it is less effective than attending lectures physically on campus. The following quote from a student illustrates dissatisfaction with online education: “*Online education is jerk. people don't see is jerk. Sitting behind my monitor 10 hours a day is jerk*.”

The third recurring theme is stress and lack of motivation. This mainly appears in the sentiment analysis, where a large portion of the responses received a negative sentiment score. Many students describe feelings such as stress, fatigue, or a lack of motivation. Deadlines and exams are often mentioned as sources of pressure. In addition, many students state that spending long hours behind a screen reduces their motivation. One student expressed this as follows: “*I don't know why it is but I am less motivated since the corona crisis.*”

These three needs were selected because they appeared repeatedly across different analysis methods. Both the word frequency analysis and the topic modelling show that themes such as studying from home, online education, and social contact are central in the dataset. In addition, the sentiment analysis confirms that many students share negative experiences related to these themes.

Solution description

Based on the students' responses and our analysis of them, we are proposing a solution to TU Delft in the form of a Social Study Hub. This solution combines a digital platform with physical study spaces that encourage students to study together and, in doing so, meet new people. This solution aims to address three key needs identified through the text analysis: a lack of social interaction among students, difficulties with online learning, and a significant drop in motivation during the COVID-19 pandemic. These needs emerged from several analytical methods: the word frequency analysis revealed terms such as 'home' and 'online'; the topic modelling analysis, on the other hand, identified clusters centred on 'contact' and 'social'; and the sentiment analysis concluded that negative experiences predominated.

The TUD Social Study Hub consists of three components. The first component is a physical study environment on campus where students can study together in small groups. This environment offers spaces for group work, quiet areas, and places where students can attend online lectures together. This allows students to participate in online education without having to do so in isolation at home. The physical study environment serves as a meeting place for students who have been matched through the platform, but also as an open study space where students can meet spontaneously. A clear separation ensures the hub remains accessible and functional for all students, while the digital platform encourages social interaction. Studying together in a physical environment can contribute to more structure in students' lives and greater social interaction. Studying together in a physical environment can help bring more structure to students' lives and foster greater social interaction. This section therefore directly addresses the lack of social interaction identified in the topic modelling results. Furthermore, words such as "social" and "contact" frequently appeared together.

The second component is an online platform where students sign up to study together. Here, students can indicate, for example, when they want to study, which course they are taking, and whether they prefer to work in a group or individually. Using this information, the system can automatically match students who are taking similar courses or want to study at the same time. This lowers the barrier to forming study groups, where students can potentially make new friends. This design choice is based on the difficulty students expressed in making connections whilst undertaking distance learning, as this is reflected both in the negative sentiment scores and the recurring mentions of isolation in the data.

The third component is a matching feature that allows students to discover and select study partners themselves, similar to the matchmaking concepts that are very popular these days. Students can view profiles of other students with similar (academic) interests and decide for themselves with whom they want to study or connect. This component gives students a sense of autonomy in building their social and academic network, in contrast to component 2, where the system holds the autonomy. This can be particularly valuable for students who are new to Delft and have not yet established many social connections. The preferences of students identified in the data address this variation in this section, where some students have a greater need for structured support, whilst others, conversely, place greater value on greater autonomy.

These findings align directly with the results of the text analysis. In the topic modelling, words such as "contact", "fellow" and "social" frequently appeared together, indicating a need for more social interaction. This was also reflected in student responses, such as: "The lack of social contacts, I'm new to Delft myself and haven't been able to meet new people yet." Words such as "online", "home", and "lecture" also appeared regularly in the analysis, suggesting that many students are struggling with prolonged online study from home. By combining physical meeting places with digital matching, the TUD Social Study Hub can alleviate these issues and give students a nudge to work on their social study environment.

Finally, the success of these components can be measured by reanalysing future surveys using the same text processing techniques. If the solution is effective, we expect words such as “lonely”, “stress”, or “lack of contact” to appear less frequently in students’ responses, while words such as “support”, “community”, “together” or “study group” will appear more often. This would indicate that student well-being has improved thanks to the TUD Social Study Hub.

Conclusion

In this report, we looked at what students needed during COVID. We did this by using computer tools to process text. First, we cleaned up the text, then we counted how often certain words appeared. We also checked how people felt based on their words and figured out the main topics they were talking about. It turns out students are mostly having trouble with not getting enough social time, finding online learning tough, and feeling less motivated and generally not as good. So, with all this in mind, we came up with the TUD Social Study Hub. It's basically a mix of real, physical places to study and an online tool that helps people find others to work with, fostering teamwork and social ties. It covers the main things, but it's not perfect. Since the responses in the dataset have been translated, we might misunderstand them, and this could make our sentiment analysis less accurate. Sometimes, tools like VADER might get the wrong idea, especially when a sentence sounds positive but actually means something negative. We also need to think about ethics. Things like privacy and whether the automated text analysis might have biases are important. So, this system should be more of a help than something that makes all the decisions on its own.

Reflection on feedback received and improvements

Based on the feedback received on Assignment 2, the report has been improved and refined in a number of areas.

- For Learning Goal 1, the pre-processing has been explained in greater detail. The steps are now explained more clearly and in a structured manner, as are our code and some additional explanation of why pre-processing is important for machine learning. This has made the process slightly more transparent and better justified.
- For Learning Goal 2, attention has been paid to the rationale behind the choices made. This applies to both the sentiment analysis (VADER) and the topic modelling, where it has been clearly explained why these methods were chosen and which parameters were used for them. We have also made it clearer why a combination of different analysis methods was used.
- Finally, in Learning Goal 3, the reasoning has been further developed, as the link between data analysis, insights and design choices has been made clearer. For instance, the solution description for each component explains how it arose from the results obtained from the analysis.

Overall, the report has therefore been improved by providing clearer steps, stronger justification and more detail behind the choices made, thereby better explaining the analysis and the design.